

Introduction to Fermi acceleration problem in dynamical systems

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Based on joint work with Jing Zhou

The original Fermi-Ulam model

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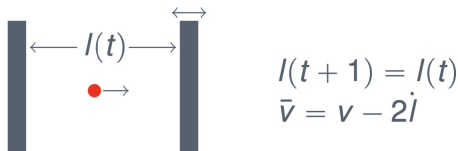


Figure: The Fermi-Ulam model

Ulam's reformulation and conjecture

Conjecture (Ulam)

There exists escaping orbits, namely the escaping set $E \neq \emptyset$.

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Escaping orbits $E = \{\limsup v_n = +\infty\}$

Bounded orbits $B = \{\exists M \text{ s.t. } |v_n| \leq M\}$

Oscillatory orbits $O = \{\limsup v_n = +\infty, \liminf v_n = 0\}$

Elimination by KAM theory

Theorem (Pustyl'nikov 1983; Laederich, Levi 1991)

If $I(t) \in C^5$, then $E = \emptyset$.

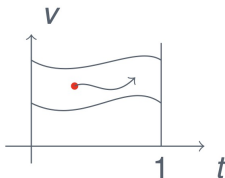


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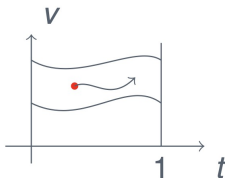


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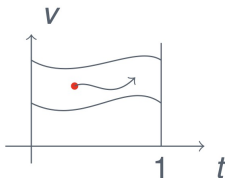


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- Zharnitsky (2000): KAM stability in quasi-periodic motions with Diophantine frequencies
- Kunze, Ortega (2018): $m(E) = 0$ with rationally independent frequencies

Attempts of Creating Acceleration with periodic functions

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- Increase dimension: **resonant billiard with moving slit(s)**

Attempts of Detecting Acceleration: non-smoothness

Zharnitsky (1998): linearly escaping orbit in a piecewise linear model



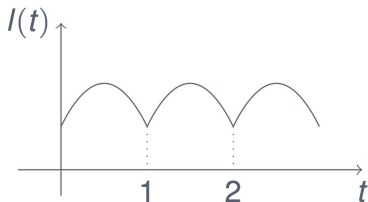
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Dolgopyat, de Simoi (2012)

- Elliptic: $m(E) = \infty, m(B) = \infty$
- Hyperbolic: $m(E) = 0, HD(E) = 2$



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- Dolgopyat 2008:
 - (1) $\alpha > 1$, $\alpha \neq 2$, $I(t)$ smooth, then $E = \emptyset$
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- Ortega 2002: $\alpha = 2$ + resonance, conditions for existence of E .

Bouncing ball in gravity

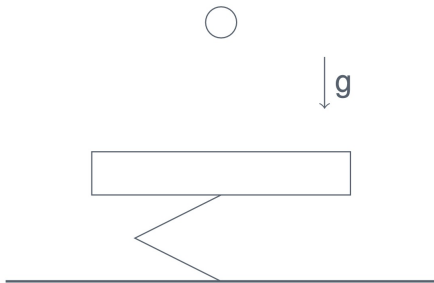


Figure: Bouncing pingpong in gravity field

$$f(t+1) = f(t), f \in C^3(0,1)$$
$$\ddot{f} > 0 \text{ for all } t, \text{ or } \ddot{f} < -g \text{ for all } t.$$

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- Rom-Kedar, Turaev, Shah 2010:
 - (1) Billiard with moving slit
 - (2) $E(\log \frac{\bar{v}}{v})$ matches the numerics in non-resonant cases
 - (3) exponential rate is significantly higher in resonant cases

A trapping billiard

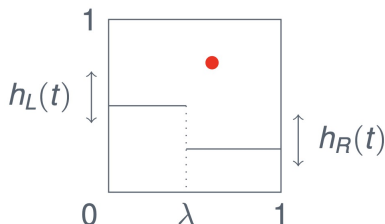


Figure: A Square Billiard with Moving Platforms

Zhou (2020):

- Almost sure exponential acceleration for high energy orbits
- With extra hyperbolicity, waiting time estimate

A switching billiard

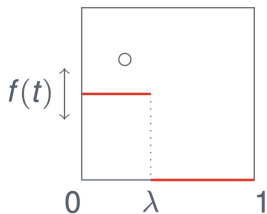


Figure: A switching billiard

Karagulyan-Zhou 2022: exponential Fermi acceleration.

Dispersing Fermi-Ulam models

Dolgopyat, de Simoi (2022-): show ergodicity, hence almost every orbit is oscillatory

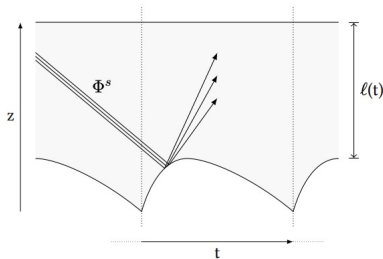


Figure: Dynamics of a dispersing Fermi-Ulam model

The dynamics behaves similarly to that of dispersing billiards (Lorentz gas) in $\mathbb{R}^2$.

What we consider

We consider piecewise linear functions:



Figure: Ulam model

$$\text{Here, } I(t) = \begin{cases} \frac{\sqrt{2}}{2}(2\{t\} + 1), & \{t\} \in (0, 1/2) \\ 2\sqrt{2} - \frac{\sqrt{2}}{2}(2\{t\} + 1), & \{t\} \in (1/2, 1) \end{cases}.$$

Theorem (Ulam model, the baby case, 2023)

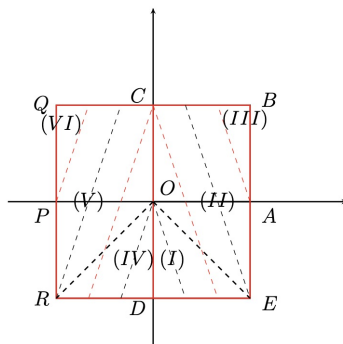
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Main result

We assume the function $I(t)$ is periodic with period 2, piecewise linear, and has two critical points $0, \alpha$. Let $M := \max\{I(t)\}$, $L := \min\{I(t)\}$. WLOG assume $ML = 1$.

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Main result, continuing

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The collision map does not project to $\mathbb{T}^2$,

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The collision map does not project to $\mathbb{T}^2$, it preserves certain invariant curves, restricted to which it is essentially an interval exchange transformation.

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Proposition

The collision map is a composition of the following two maps:

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$$P_\alpha^0 : \begin{cases} \bar{\tau} = \tau - (2 - \alpha)I \pmod{1} \\ \bar{I} = I - MK(\bar{\tau} - 1) \end{cases}$$

where $K = (M - L)(1/\alpha + 1/(2 - \alpha))$.

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The collision map restricted to each invariant curve (x_0, l_0) , is equivalent to the map $\mathcal{P} = f \circ h \circ f \circ g$

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Question

For a typical dynamical system (T_t, X, μ) , $\phi : X \rightarrow \mathbb{R}$. What happens if $I(t) = \phi(T_t x)$?

Thank you for your attention.